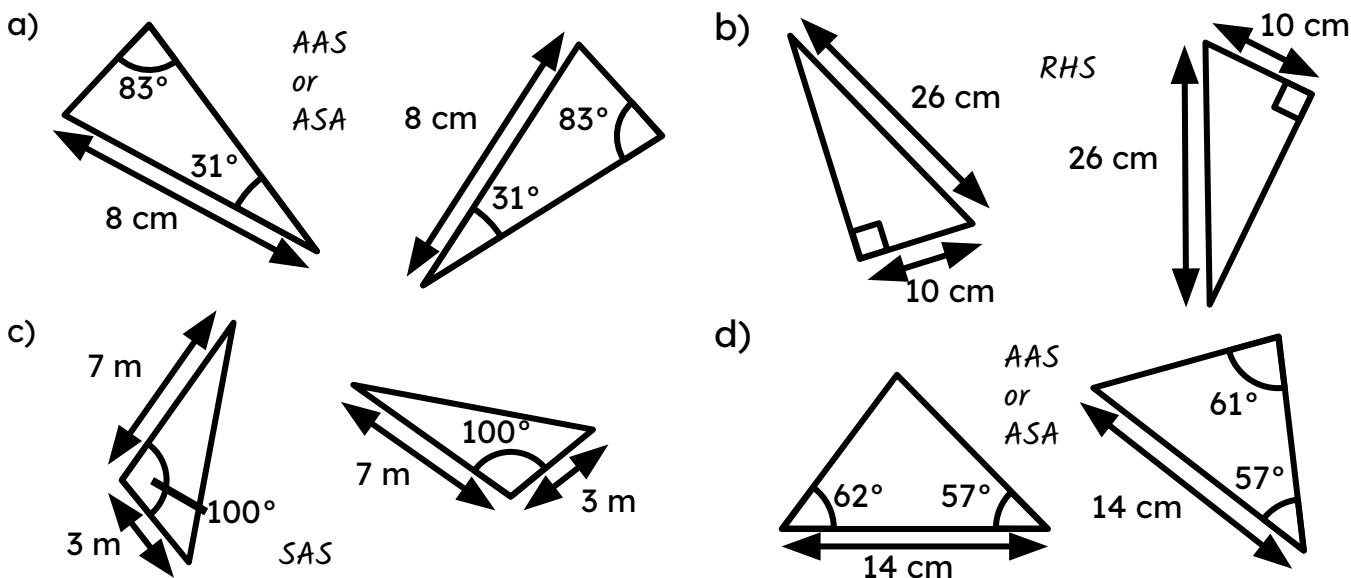




Applying the criteria for congruence

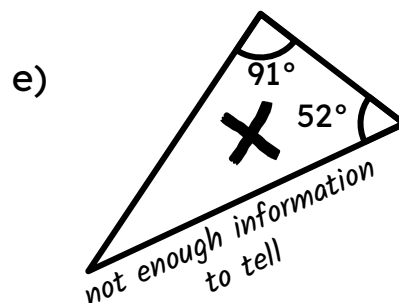
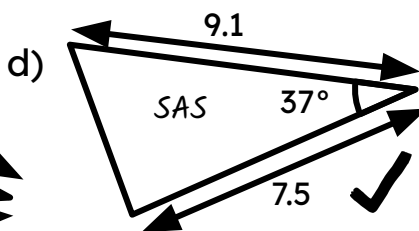
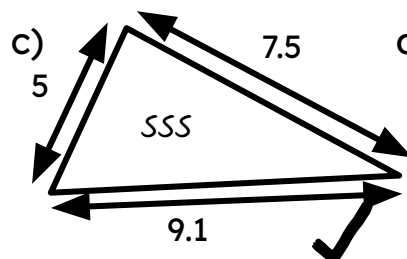
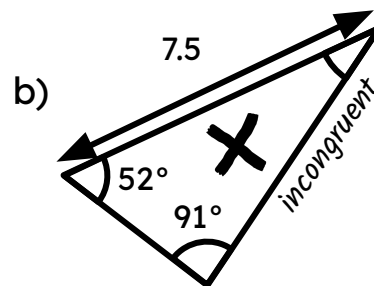
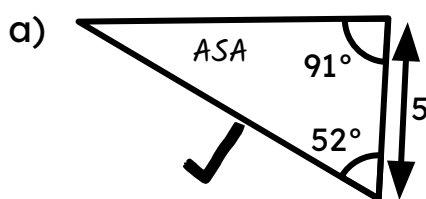
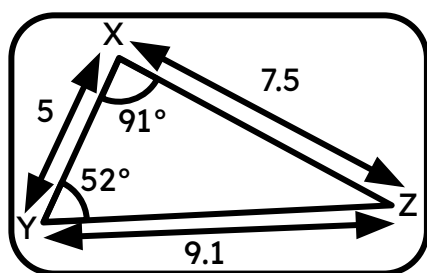
Task A: Choosing the correct criteria

1) What criteria proves each pair to be congruent to each other?

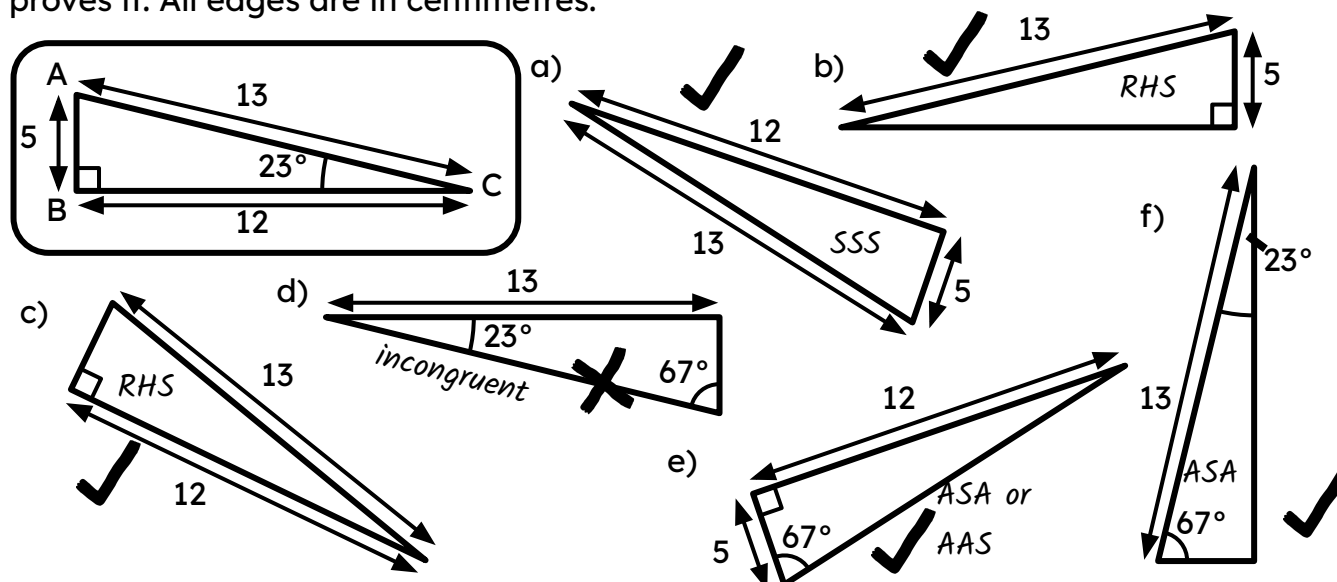


Task B: Congruent or not?

1) Decide which of the triangles are congruent to triangle XYZ, giving the criteria that proves it. All edges are in centimetres.



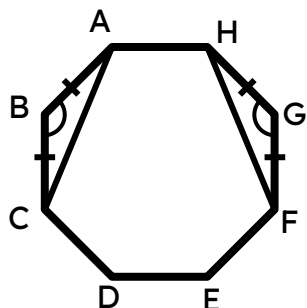
2) Decide which of the triangles are congruent to triangle ABC. State the criteria that proves it. All edges are in centimetres.



Task C: Hidden congruence

1) ABCDEFGH is a regular octagon.

a) Prove triangles ABC and triangle HGF are congruent.



$AB = BC = HG = GF$ as the octagon is regular

$\angle ABC = \angle HGF$ as they are interior angles of the regular octagon

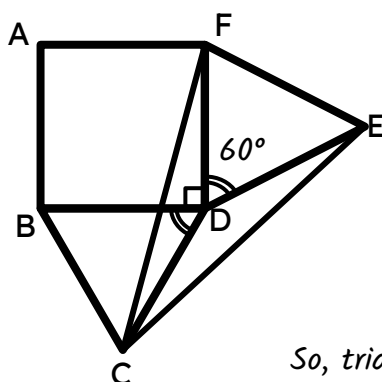
Triangle ABC is congruent to triangle HGF by SAS.

b) What other triangles are congruent to ABC?

Triangles BCD, CDE, DEF, EFG, GHA and HAB are congruent to ABC

2) ABDF is a square. BCD and DEF are equilateral triangles.

Prove triangles CDE and FDC are congruent.



CD is a common edge to both triangles

$DF = DE$ as they are edges of the same equilateral triangle

$$\angle CDF = 90 + 60 = 150^\circ$$

$$\angle CDE = 360 - (60 + 90 + 60) = 150^\circ$$

Therefore, $\angle CDF = \angle CDE$

So, triangle CDE is congruent to triangle FDC by SAS.

